1.2 Earth's Life Support Systems

This topic examines the water and carbon cycles as fundamental Earth systems that sustain life. It explores how these cycles operate at scales from local to global, how they are interconnected, and how human activity is disrupting their natural balance with profound consequences for climate, ecosystems, and human societies.

The Water Cycle (Hydrological Cycle)

Hydrological cycle: The continuous movement of water between the atmosphere, land surface, oceans, and subsurface stores, driven primarily by solar energy (evaporation) and gravity (precipitation, runoff, percolation). At the **global scale**, it is a **closed system** — no water is created or destroyed; the total amount of water on Earth remains constant (~1.4 billion km^3). At the **local/drainage basin scale**, it operates as an **open system** with inputs (precipitation) and outputs (river discharge, evapotranspiration) crossing the system boundary (the watershed).

Key Stores

Store	% of Global Water	Approximate Volume	Residence Time
Oceans	96.5%	~1,335 million km^3	~3,200 years
Ice caps, glaciers, permanent snow	1.74%	~26 million km^3	~10,000 years (glaciers); 1 million+ years (Antarctic ice sheet)
Groundwater (total)	1.69%	~10.5 million km^3	Hundreds to thousands of years
Freshwater lakes	0.013%	~91,000 km^3	~17 years

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Soil moisture	0.001%	~16,500 km^3	Weeks to months
Atmosphere (water vapour)	0.001%	~12,900 km^3	~9 days
Rivers	0.0002%	~2,120 km^3	~16 days
Biosphere (living organisms)	0.0001%	~1,120 km^3	~1 week

Exam Tip: Only about **2.5%** of all water on Earth is freshwater, and most of that is locked in ice caps and glaciers (~68.7% of freshwater) or deep groundwater (~30.1% of freshwater). Only about **0.3%** of freshwater is readily accessible in rivers, lakes, and shallow groundwater. This makes freshwater a precious and unevenly distributed resource. Use these statistics to support arguments about water security and resource conflicts.

Key Flows (Transfers and Transformations)

- **Evaporation:** Liquid water converts to water vapour using solar energy (latent heat of vaporisation). Greatest over warm tropical oceans. Rate increases with temperature, wind speed, and low humidity. Global evaporation: ~505,000 km^3/year
- **Transpiration:** Water loss from plants through open stomata on leaf surfaces. A single large tree can transpire 100+ litres per day. Accounts for ~10% of atmospheric moisture. A major flux in forested areas
- **Evapotranspiration:** The combined water loss from evaporation (from surfaces) and transpiration (from plants). Measured using lysimeters. Potential evapotranspiration (PET) is the amount that would occur if water were unlimited; actual evapotranspiration is often lower in arid areas
- **Condensation:** Water vapour cools (reaches dew point) and converts to liquid droplets around condensation nuclei (dust, salt, pollen). Forms clouds, fog, and dew
- **Precipitation:** Water falling from the atmosphere as rain, snow, sleet, or hail when cloud droplets grow heavy enough to fall. Global precipitation: ~505,000 km^3/year (balanced with evaporation in the long term)

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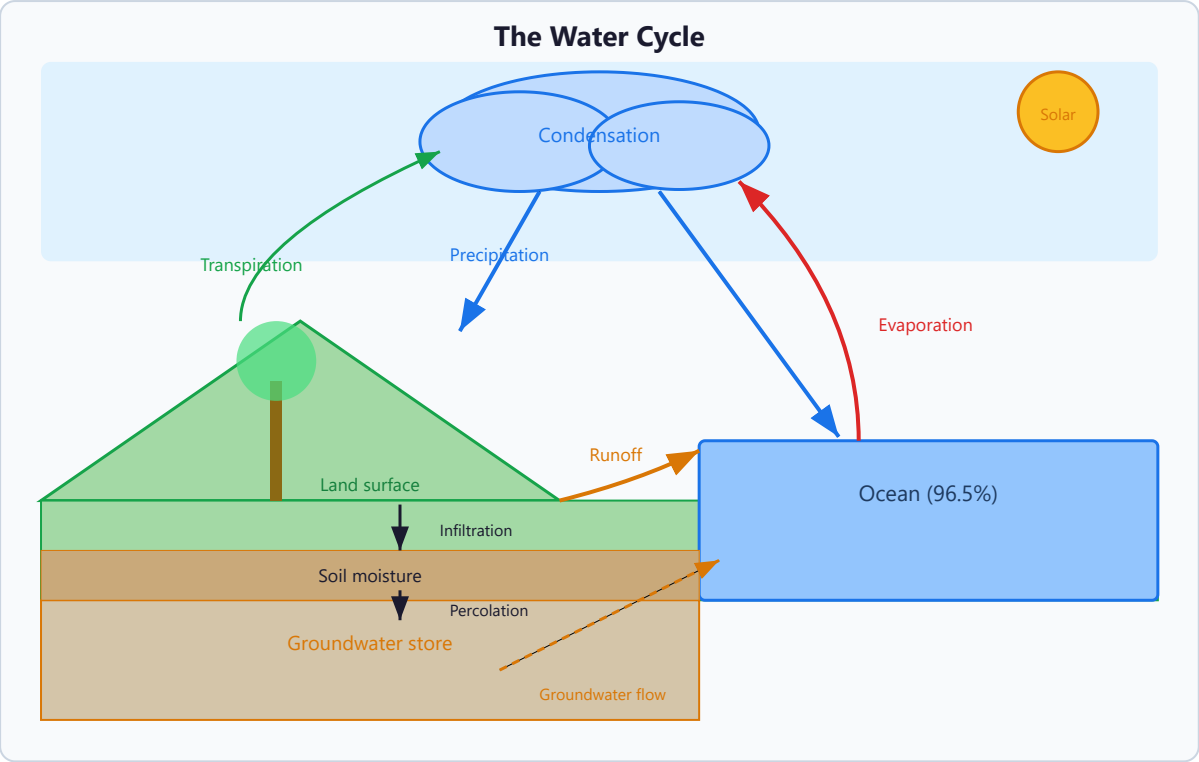
- **Interception:** Precipitation caught by vegetation canopy or built surfaces before reaching the ground. Can account for 10–40% of rainfall in forested areas. Intercepted water evaporates directly back to the atmosphere or reaches the ground via stemflow or throughfall
- **Infiltration:** Water soaking into the soil from the surface, driven by gravity and capillary action. Rate depends on soil porosity, permeability, antecedent moisture, vegetation cover, and land use. Infiltration capacity is the maximum rate at which soil can absorb water
- **Percolation:** Deeper, slower movement of water through soil and rock into the groundwater store (aquifer). Occurs through permeable rocks like chalk, sandstone, and limestone
- **Throughflow:** Lateral (sideways) movement of water through the soil layer towards rivers and streams. Slower than surface runoff but faster than groundwater flow. Often follows natural pipes and macropores in the soil
- **Groundwater flow:** Very slow movement of water through saturated permeable rock (aquifer) towards rivers, lakes, or the sea. The water table marks the upper surface of the saturated zone. Groundwater feeds rivers as **baseflow**, maintaining river levels during dry periods
- **Surface runoff (overland flow):** Water flowing over the ground surface directly into rivers. Occurs when rainfall intensity exceeds infiltration capacity (**Hortonian overland flow**) or when the soil is fully saturated (**saturation overland flow**). Fastest transfer — responds quickly to rainfall, contributing to flood peaks
- **Stemflow:** Water running down plant stems and trunks to the ground after interception
- **Channel flow:** Water flowing within a river channel towards the sea. Velocity depends on gradient, channel shape, roughness, and discharge. Measured in **cumecs** (m^3/s)



Drainage Basin Hydrology

Drainage basin: The area of land drained by a river and all its tributaries. It is an **open system** bounded by a **watershed** (ridge of high land forming the boundary). Water enters as precipitation and leaves via the river mouth (discharge) and evapotranspiration. The drainage basin is the fundamental unit for studying the water cycle at a local/regional scale.

Water balance equation:

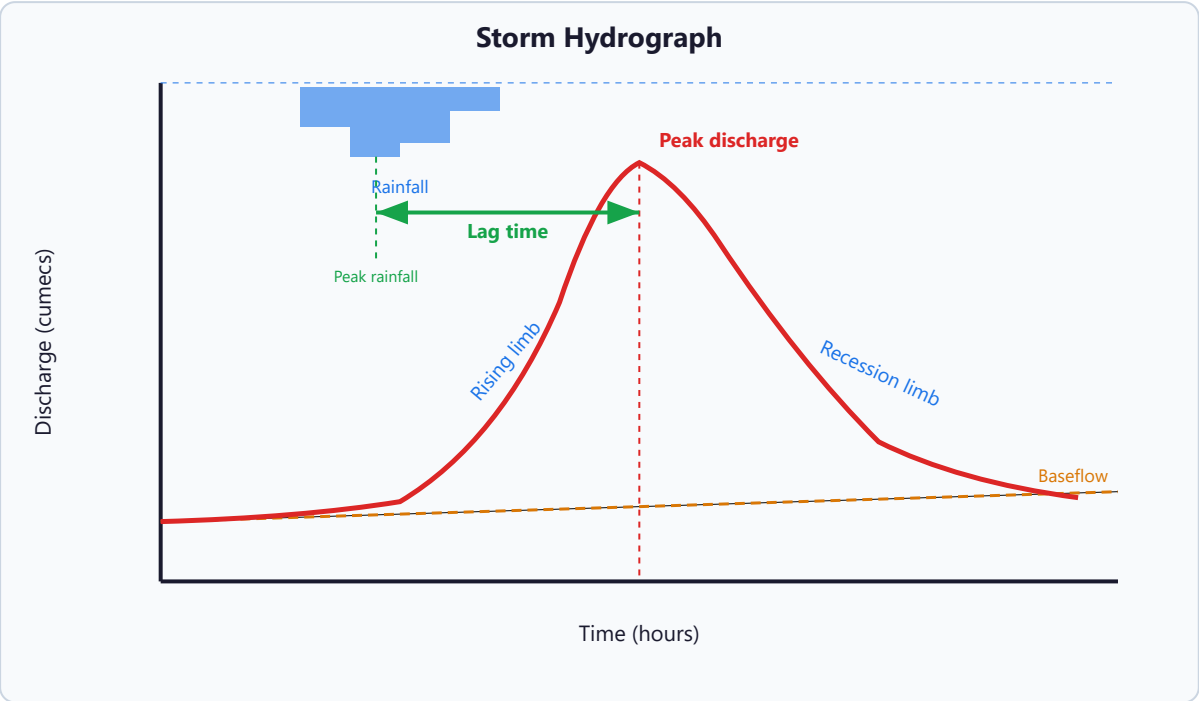
$$P = Q + E \pm \Delta S$$

Where P = precipitation, Q = discharge (river output), E = evapotranspiration, ΔS = change in storage (soil moisture, groundwater, lakes). If ΔS is positive, the basin is gaining water (wet season); if negative, it is losing stored water (dry season).

Storm Hydrographs

Storm hydrograph: A graph showing how **river discharge** changes over time in response to a rainfall event. Key features: **peak discharge** (maximum flow), **lag time** (time between peak rainfall and peak discharge), **rising limb** (increasing discharge as rainfall reaches the river), **falling/recession**

limb (declining discharge as water drains from stores), and **baseflow** (the steady contribution from groundwater seeping into the river).



Factors Affecting Storm Hydrograph Shape

A **flashy hydrograph** (short lag time, high peak discharge, steep rising limb) indicates rapid water transfer to the river, increasing flood risk. A **flat hydrograph** (long lag time, low peak, gentle rising limb) indicates water is being stored and slowly released.

Factor	Flashy Hydrograph (High Flood Risk)	Flat Hydrograph (Low Flood Risk)
Soil/rock type	Impermeable clay or thin soils (water cannot infiltrate, so runs off)	Permeable sand, chalk, or thick soils (water infiltrates and percolates slowly)
Land use	Urbanised: concrete, tarmac, drains (impermeable surfaces, rapid channel transfer)	Forested: high interception, infiltration, transpiration (water stored in vegetation and soil)
Drainage density	High density of streams → water enters channels quickly	Low density → water travels slowly through soil as throughflow

Rainfall intensity	Heavy, intense bursts (exceeds infiltration capacity → surface runoff)	Gentle, prolonged rainfall (infiltrates into soil)
Slope gradient	Steep slopes → fast surface runoff, limited infiltration time	Gentle slopes → water has time to infiltrate
Antecedent moisture	Soil already saturated from previous rain (no more absorption capacity)	Dry soil (high infiltration capacity, absorbs rainfall)
Basin shape	Circular (tributaries deliver water to the main channel simultaneously)	Elongated (tributaries deliver water at staggered times, spreading the peak)
Season	Winter: saturated soil, low transpiration, deciduous trees bare	Summer: higher evapotranspiration, drier soil, leafy canopy intercepts rain

Exam Tip: When analysing a hydrograph, always explain the link between the physical characteristic and the hydrological process. For example: "Urbanisation increases the peak discharge and reduces lag time because impermeable surfaces (concrete, tarmac) prevent infiltration, so precipitation rapidly becomes surface runoff, entering storm drains and reaching the river channel very quickly." Do not just state the factor — explain the mechanism.

The Carbon Cycle

Carbon cycle: The biogeochemical cycle by which carbon is exchanged between the atmosphere, oceans, biosphere (living organisms), pedosphere (soils), and lithosphere (rocks and fossil fuels). Carbon is the fundamental element of all organic life and its movement between these stores — through processes like photosynthesis, respiration, combustion, weathering, and ocean absorption — regulates Earth's climate through the greenhouse effect. The carbon cycle operates on timescales from seconds (respiration) to millions of years (rock formation).

Carbon Stores

Store	Approximate Amount (GtC)	Residence Time	Key Detail
Lithosphere (sedimentary rocks + fossil fuels)	~100,000,000	Millions of years	By far the largest store. Includes limestone ($CaCO_3$), coal, oil, gas. Carbon enters via sedimentation and burial; exits via volcanic outgassing and human extraction/combustion
Deep ocean	~38,000	1,000s of years	Dissolved inorganic carbon (bicarbonate/carbonate ions), dissolved organic carbon, and particulate matter. The thermohaline circulation (oceanic conveyor belt) transports carbon to the deep ocean
Soil organic matter	~1,500–2,400	Decades to centuries	Humus from decomposed organisms. Contains more carbon than the atmosphere and terrestrial vegetation combined. Peatlands alone store ~600 GtC. Soil carbon is vulnerable to release through land-use change, drainage, and warming
Surface ocean	~900	Years to decades	CO_2 dissolves at the surface (governed by Henry's Law: colder water absorbs more). Phytoplankton fix carbon via photosynthesis. The ocean is currently a net carbon sink (~2.5 GtC/year)
Atmosphere	~830 (and rising)	~5 years for a CO_2 molecule	Mainly CO_2 (~420 ppm in 2024, up from ~280 ppm pre-industrial) and CH_4 (~1,900 ppb). Small store but critical for regulating temperature via the greenhouse effect
Terrestrial vegetation	~560	Years to decades	Carbon fixed by photosynthesis and stored in wood, leaves, roots. Tropical rainforests are the largest terrestrial carbon store per unit area. Temperate and boreal forests store significant carbon in slow-growing wood

Carbon Fluxes (Transfers)

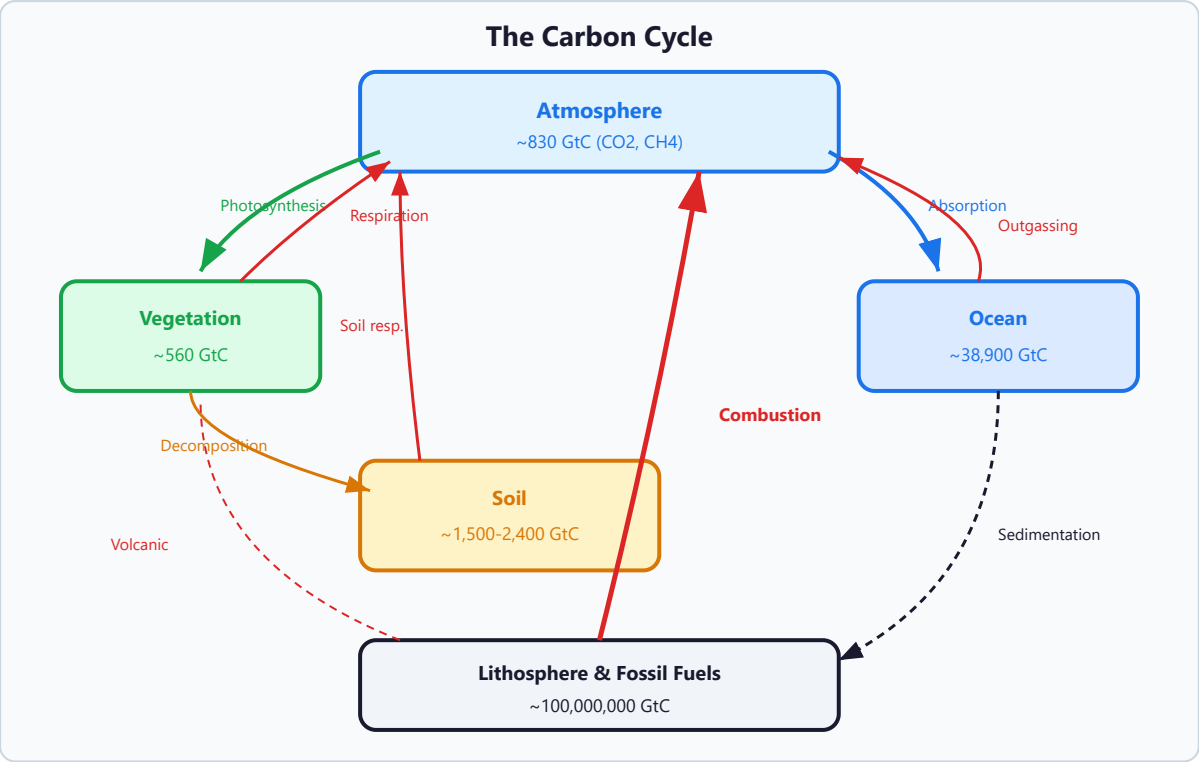
Biological fluxes:

- **Photosynthesis:** Plants, algae, and cyanobacteria absorb CO_2 from the atmosphere and convert it to organic carbon compounds (glucose) using solar energy. $\mathbf{6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2}$. This is the primary mechanism transferring carbon from the atmosphere to the biosphere. Global terrestrial photosynthesis fixes ~120 GtC/year
- **Respiration:** All living organisms (plants, animals, microbes) break down organic carbon compounds to release energy, producing CO_2 as a waste product. The reverse of photosynthesis. Returns ~119 GtC/year to the atmosphere from terrestrial sources (approximately balanced with photosynthesis before human interference)
- **Decomposition:** Dead organic matter (detritus) is broken down by bacteria and fungi (decomposers), releasing CO_2 (aerobic decomposition) or CH_4 (anaerobic decomposition in waterlogged conditions like wetlands and rice paddies). Rate depends on temperature (faster when warm), moisture, oxygen availability, and the chemical composition of the material

Geological and chemical fluxes:

- **Combustion of fossil fuels:** Burning coal, oil, and natural gas releases carbon stored over millions of years. This is the largest human-caused carbon flux (~9.5 GtC/year and rising). It has increased atmospheric CO_2 by ~50% since the Industrial Revolution
- **Chemical weathering:** Carbonic acid (formed from CO_2 dissolving in rainwater) reacts with silicate and carbite rocks, drawing down atmospheric CO_2. Very slow but significant over geological timescales (millions of years). Acts as a natural negative feedback on atmospheric CO_2
- **Volcanic outgassing:** CO_2 and other gases released from Earth's interior during volcanic eruptions and through hydrothermal vents. Approximately 0.15–0.26 GtC/year — small compared to human emissions but important over geological time
- **Ocean absorption and outgassing:** CO_2 dissolves in surface seawater (governed by Henry's Law — cold water absorbs more than warm water). CO_2 reacts with water to form carbonic acid (H_2CO_3), contributing to **ocean acidification** (pH has fallen by ~0.1 units since pre-industrial, a 26% increase in acidity). The ocean is currently absorbing ~2.5 GtC/year (a critical carbon sink), but its capacity may decrease as waters warm
- **Sedimentation and lithification:** Dead marine organisms with calcium carbonate shells (foraminifera, corals, molluscs) sink to the ocean floor. Over millions of years, these sediments are compressed and

lithified into limestone and chalk, locking carbon away in the lithosphere for geological timescales



Interconnections Between Water and Carbon Cycles

The water and carbon cycles are deeply and fundamentally linked — disrupting one inevitably affects the other. Understanding these interconnections is essential for OCR examiners:

- **Vegetation links both cycles:** Plants need water for photosynthesis (which sequesters atmospheric carbon). Transpiration drives the local water cycle by returning water vapour to the atmosphere. Forests are simultaneously **carbon sinks** (absorbing CO_2) and **drivers of local water cycling** (generating atmospheric moisture that falls as "recycled" precipitation). Deforestation disrupts both cycles simultaneously
- **Oceans link both cycles:** The ocean is both the largest water store (96.5%) and a major carbon sink (~2.5 GtC/year). CO_2 dissolves more readily in cold water (Henry's Law), so cold polar waters are important CO_2 sinks. The thermohaline circulation transports dissolved carbon to the deep ocean (the "biological pump" and "solubility pump")
- **Soil links both cycles:** Soil stores both water (soil moisture) and carbon (humus, organic matter). **Waterlogged conditions** (anaerobic) dramatically slow decomposition, preserving soil carbon — this is

how **peat bogs** form (accumulating carbon over thousands of years). Draining peatlands exposes the organic matter to oxygen, rapidly releasing stored carbon as CO_2

- **Climate links both cycles:** The carbon cycle regulates temperature via the greenhouse effect, which in turn drives evaporation rates and precipitation patterns in the water cycle. A warmer atmosphere holds more water vapour (~7% more per degree C — Clausius-Clapeyron relation), intensifying the hydrological cycle (more intense rainfall events, greater flood risk)
- **Chemical weathering links both cycles:** Requires water (as carbonic acid: $CO_2 + H_2O \rightarrow H_2CO_3$) to dissolve silicate and carbonate rocks, drawing down atmospheric CO_2. More rainfall → more weathering → more CO_2 removal (a slow negative feedback on warming)

Exam Tip: OCR examiners frequently ask about **interconnections** between the two cycles. Always link processes across both systems. For example: "Deforestation reduces photosynthesis (carbon cycle: less CO_2 sequestered) AND reduces interception and transpiration (water cycle: less evapotranspiration, more surface runoff, increased flood risk, reduced local rainfall through loss of moisture recycling). This demonstrates how a single human action disrupts both cycles simultaneously, with cascading consequences." This kind of synoptic thinking scores very highly.

Climate Change

The Greenhouse Effect

The **natural greenhouse effect** is essential for life: greenhouse gases (CO_2, CH_4, N_2O, water vapour, ozone) in the atmosphere absorb and re-emit outgoing longwave (infrared) radiation from Earth's surface, trapping heat and keeping the global mean temperature at about **15 degrees C** (without them it would be approximately **−18 degrees C**). The **enhanced greenhouse effect** is the additional warming caused by human-induced increases in greenhouse gas concentrations, particularly CO_2 from fossil fuel combustion and CH_4 from agriculture.

Evidence for Recent Climate Change

Evidence Type	Detail	Significance

Ice cores	Air bubbles trapped in Antarctic ice (Vostok, EPICA Dome C) provide a record of atmospheric CO_2 and temperature going back ~800,000 years	Show a strong correlation between CO_2 and temperature throughout glacial-interglacial cycles. Current CO_2 (420 ppm) is far higher than any point in the 800,000-year record (natural range: 180–280 ppm)
Instrumental records	Thermometer records since ~1850 show global mean surface temperature has risen by approximately **1.1 degrees C** above pre-industrial levels (as of 2023). 2023 was the warmest year on record	The rate of warming is unprecedented in geological context. Most warming has occurred since 1970
Sea-level rise	Global mean sea level has risen ~20 cm since 1900 and is currently rising at ~3.6 mm/year (accelerating)	Caused by thermal expansion of ocean water (~42%) and melting of ice sheets and glaciers (~58%). Threatens low-lying islands and coastal cities
Glacial retreat	Most mountain glaciers worldwide are shrinking. Glacier National Park (USA) had 150 glaciers in 1850; fewer than 26 remain. The Greenland ice sheet is losing ~280 Gt of ice per year	Visual, measurable evidence of warming. Glaciers respond directly to temperature changes
Arctic sea ice decline	September Arctic sea ice extent has declined by ~13% per decade since satellite records began (1979). 2012 and 2020 were record lows	Accelerates warming through ice-albedo positive feedback. Opens shipping routes but threatens Arctic ecosystems and indigenous communities
Ocean acidification	Ocean pH has dropped by ~0.1 units since pre-industrial times (a 26% increase in hydrogen ion concentration)	Threatens calcifying organisms (corals, shellfish, plankton) that cannot build shells/skeletons in more acidic water. Impacts marine food chains
Phenological changes	Earlier spring flowering, earlier bird migration, shifts in species ranges towards poles and higher altitudes	Biological evidence that ecosystems are responding to warming. Can cause mismatches (e.g. birds arriving before insect food is available)

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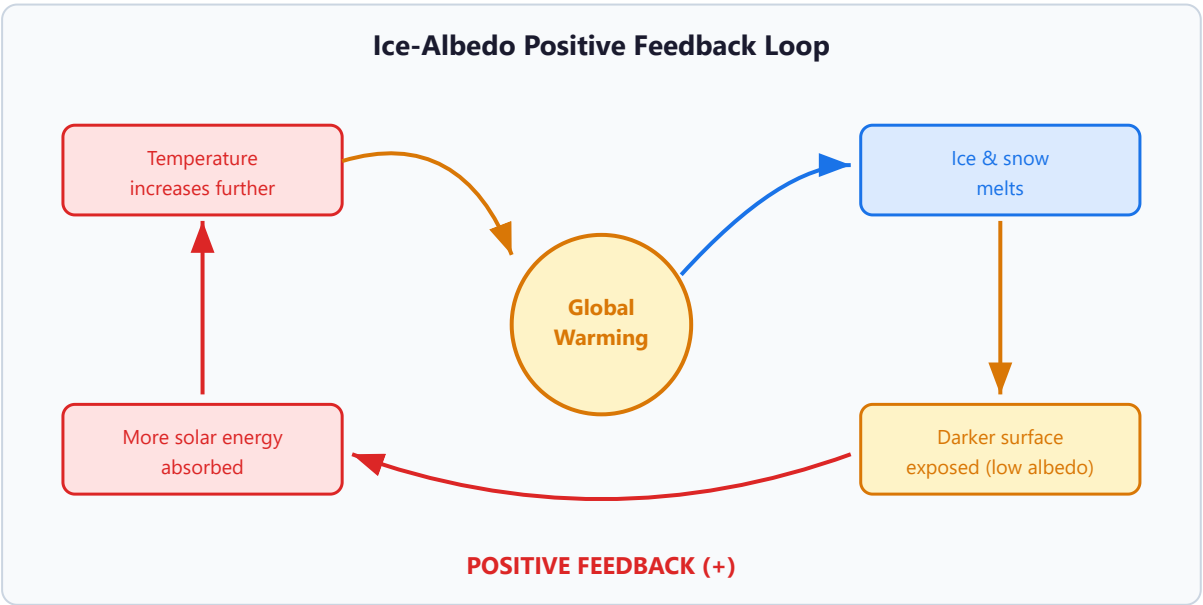
Causes of Climate Change

Natural Causes	Human (Anthropogenic) Causes
Milankovitch cycles: Periodic changes in Earth's orbital eccentricity (~100,000 years), axial tilt (~41,000 years), and precession (~26,000 years) alter the distribution and intensity of solar radiation reaching Earth. Drive glacial-interglacial cycles	**Fossil fuel combustion:** Burning coal, oil, and gas for energy, transport, and industry releases CO_2. Responsible for ~75% of anthropogenic greenhouse gas emissions. CO_2 has risen from 280 ppm to >420 ppm
Solar output variations: Small (~0.1%) fluctuations in solar energy output over the 11-year sunspot cycle. Cannot explain the magnitude of recent warming	**Deforestation:** Removing forests eliminates a carbon sink (reduced photosynthesis) and releases stored carbon when trees are burned or decompose. Responsible for ~10% of anthropogenic CO_2 emissions
Volcanic eruptions: Large eruptions inject SO_2 aerosols into the stratosphere, reflecting sunlight and causing temporary cooling (1–3 years). Also emit small amounts of CO_2	**Agriculture:** CH_4 from livestock digestion (enteric fermentation) and rice paddies; N_2O from synthetic fertilisers; CO_2 from machinery and land clearance. Agriculture is responsible for ~10–12% of global emissions
Ocean circulation changes: El Nino/La Nina cycles, thermohaline circulation variations affect heat distribution and CO_2 exchange	**Industrial processes:** Cement production (heating limestone releases CO_2: $CaCO_3 \rightarrow CaO + CO_2$), steel manufacturing, chemical production. Cement alone accounts for ~8% of global CO_2 emissions

Feedback Mechanisms in Climate Change

Feedback	Type	Mechanism
Ice-albedo feedback	Positive	Warming melts ice and snow → darker land/ocean surface exposed → lower albedo → absorbs more solar radiation → more warming → more ice melts. Particularly powerful in the Arctic (Arctic is warming ~3x faster than the global average)

Permafrost thaw	Positive	Warming thaws permafrost (permanently frozen ground containing ~1,500 GtC in organic matter) → decomposition of previously frozen organic material releases CH_4 and CO_2 → enhanced greenhouse effect → more warming → more thaw. A potential "tipping point" — once initiated, may become irreversible
Water vapour feedback	Positive	Warming increases evaporation → more water vapour in atmosphere (water vapour is the most abundant greenhouse gas) → enhanced greenhouse effect → more warming. Amplifies CO_2-driven warming by a factor of ~2
Ocean solubility pump weakening	Positive	Warmer oceans absorb less CO_2 (solubility decreases with temperature) → more CO_2 remains in atmosphere → more warming. The ocean's role as a carbon sink may weaken over time
Cloud feedback	Mixed / uncertain	Warming increases evaporation → more clouds. **Low clouds** (cumulus) have high albedo and cool the surface (negative feedback). **High clouds** (cirrus) trap outgoing radiation and warm the surface (positive feedback). The net effect is one of the largest uncertainties in climate models
CO_2 fertilisation	Negative (limited)	Higher CO_2 → increased rate of photosynthesis → plants absorb more CO_2 → slows atmospheric CO_2 increase. However, this effect is limited by water and nutrient availability, and may saturate at higher CO_2 levels



Human Impacts on the Carbon Cycle

Deforestation

- Removes a major carbon sink (trees absorb CO_2 via photosynthesis and store it in biomass for decades to centuries)
- Burning or decomposition of felled trees releases stored carbon back to the atmosphere as CO_2
- Exposed soil dries out and oxidises, releasing additional CO_2. Erosion by rain removes carbon-rich topsoil
- Reduces transpiration, decreasing local atmospheric moisture and potentially reducing rainfall (positive feedback: less rain → forest stress → more dieback → less rain)
- Deforestation is responsible for approximately **10% of global CO_2 emissions**

Case Study: Amazon Rainforest

The Amazon basin stores approximately **150–200 billion tonnes of carbon** in its biomass and soil, making it one of the most important carbon stores on Earth. Deforestation — primarily for cattle ranching (~80% of cleared land), soy cultivation, and logging — is releasing this carbon at an alarming rate. Between 2001 and 2022, approximately **17–20%** of the original Amazon forest was lost or severely degraded. Recent studies (Gatti et al., 2021) found that parts of the **eastern Amazon have already become a net carbon source** rather than a sink, emitting more CO_2 than they absorb due to fires and deforestation. Deforestation also reduces evapotranspiration — the Amazon generates approximately **50% of its own rainfall** through moisture recycling (the "flying rivers" phenomenon). Continued deforestation risks pushing the Amazon past a **tipping point** (~20–25% deforestation) where reduced rainfall triggers irreversible large-scale dieback, converting forest to savanna and releasing catastrophic amounts of stored carbon. This would have global climate consequences.

Fossil Fuel Combustion

- Transfers carbon from the long-term geological store (where it was locked away over **millions of years**) to the atmosphere in a **geological instant** (decades)
- Atmospheric CO_2 has risen from ~280 ppm (pre-industrial, 1750) to over **420 ppm** (2024) — a 50% increase and the highest level in at least 3 million years (Pliocene epoch)

- Fossil fuel combustion is responsible for approximately **75%** of all anthropogenic greenhouse gas emissions (~36 $GtCO_2$/year or ~9.5 GtC/year)
- Fundamentally disrupts the balance between natural carbon fluxes that maintained stable atmospheric CO_2 for millennia

Human Impacts on the Water Cycle

- **Urbanisation:** Impermeable surfaces (concrete, tarmac, roofs) dramatically increase surface runoff and reduce infiltration. Storm drains channel water rapidly to rivers. Result: flashier hydrographs, increased flood frequency and magnitude, reduced groundwater recharge. Example: the area of urban land in England increased by 16% between 1990 and 2018
- **Deforestation:** Reduces interception (no canopy to catch rain) and transpiration (no trees to return moisture to atmosphere). Result: more surface runoff, increased flood risk, reduced local rainfall through loss of moisture recycling, soil erosion. The Amazon generates ~50% of its own rainfall through evapotranspiration
- **Agriculture:** Irrigation depletes groundwater stores and rivers (the Ogallala Aquifer in the US Great Plains is declining by ~1 m/year in some areas). Drainage of wetlands removes natural water storage. Ploughing increases runoff. Fertiliser runoff causes eutrophication of water bodies
- **Dam construction:** Creates new storage (reservoir), alters downstream river regime (flow pattern), traps sediment (downstream erosion), enables irrigation and hydroelectric power, can displace communities. Example: Three Gorges Dam (China) displaced 1.3 million people
- **Climate change:** Intensifies the hydrological cycle (warmer atmosphere holds ~7% more moisture per degree C). Leads to: more intense rainfall events (increased flood risk), longer dry periods (drought risk), glacial melt (temporary increase then loss of meltwater supply), rising sea levels, altered monsoon patterns

Case Study: Aral Sea, Central Asia

Once the **world's 4th largest lake** (68,000 km^2 in 1960), the Aral Sea has shrunk catastrophically since the 1960s when the Soviet Union diverted the **Amu Darya and Syr Darya rivers** to irrigate cotton fields in Uzbekistan and Kazakhstan. By 2014, the eastern basin had **completely dried up** for the first time in 600 years. The lake has lost over 90% of its volume. Consequences: (1) **Collapse of the fishing industry** — 60,000 jobs lost; the port of Moynaq is now 150 km from the water, (2) **Salinisation** — remaining water became hypersaline (100+ g/L), killing aquatic life, (3) **Dust storms**

— toxic dust containing pesticides and salt from the exposed seabed blows over surrounding populations, causing respiratory disease, kidney disease, and cancer, (4) **Climate change** — loss of the moderating influence of the lake has made local climate more extreme (hotter summers, colder winters), (5) **Desertification** of the former lake bed. The Aral Sea is one of the most dramatic examples of how **human alteration of the water cycle** can cause environmental and humanitarian catastrophe. Partial recovery of the northern Aral Sea has been achieved through the Kok-Aral Dam (funded by the World Bank), but the southern basin is considered irreversibly lost.

Management and Mitigation

Carbon Cycle Management

Strategy	Description	Evaluation
REDD+	Reducing Emissions from Deforestation and forest Degradation: pays developing countries to protect existing forests and plant new ones	Addresses ~10% of emissions. Challenges include monitoring, corruption, indigenous rights, and "carbon leakage" (deforestation shifts to unprotected areas)
Carbon capture and storage (CCS)	Captures CO_2 from power station emissions and stores it underground in depleted oil/gas reservoirs or saline aquifers	Technology works but is extremely expensive and not yet deployed at scale. Only ~40 CCS facilities worldwide. Risk of leakage from underground storage
Afforestation / reforestation	Planting new forests or restoring degraded forests to increase carbon sequestration	Effective and relatively cheap. The Bonn Challenge aims to restore 350 million hectares by 2030. However, monoculture plantations are less effective than natural forests, and trees take decades to mature as significant carbon stores
Renewable energy transition	Replacing fossil fuels with solar, wind, hydro, tidal, geothermal, and nuclear energy	Essential for deep decarbonisation. Costs have fallen dramatically (solar PV down 89% since 2010). Challenges: intermittency, energy

		storage, grid infrastructure, political resistance from fossil fuel interests
International agreements	Paris Agreement (2015): signatories commit to limiting warming to **1.5–2 degrees C** above pre-industrial levels through Nationally Determined Contributions (NDCs)	Historic agreement but NDCs are voluntary, and current pledges (if fully met) would result in ~2.5–2.8 degrees C of warming. No enforcement mechanism. US withdrew under Trump (2017–2021, rejoined 2021)

Water Cycle Management

Strategy	Description	Example
Sustainable drainage systems (SuDS)	Permeable paving, green roofs, rain gardens, retention ponds, swales — designed to mimic natural drainage and reduce surface runoff in urban areas	London Olympic Park used extensive SuDS; Sheffield Grey to Green project converted road surfaces to rain gardens, reducing flood risk and improving urban amenity
Wetland restoration	Restoring drained wetlands to act as natural flood storage, carbon sinks, and biodiversity habitats	Wicken Fen (Cambridgeshire): 100-year plan to restore 5,300 hectares of former farmland to wetland. The Great Fen project
Water conservation	Reducing demand through water meters, efficient irrigation (drip irrigation rather than spray), water recycling, leak reduction	Israel recycles 86% of its wastewater for agricultural irrigation (world leader). Thames Water loses ~24% of supply through pipe leaks
Catchment-scale management	Integrated approach managing land use, forestry, farming, and development across entire drainage basins to reduce flood risk and improve water quality	Slowing the Flow at Pickering (North Yorkshire): planting trees, installing leaky dams, and creating bunds in the upper catchment reduced peak flood discharge by 15–20%

Exam Tip: For 20-mark essays on this topic, structure your answer around: (1) **the natural cycle** (stores, flows, key processes), (2) **human disruption** of stores and flows (with specific data), (3)

consequences and feedbacks (both positive and negative), (4) **management strategies** (evaluate effectiveness, not just describe). Always include specific data and named case studies (Amazon: 150–200 GtC stored; CO_2: 280 → 420 ppm; Aral Sea: 90% volume loss). Show awareness that the **water and carbon cycles are interconnected** — disrupting one affects the other. For the highest marks, discuss **tipping points** and **feedback loops** that could push systems past the point of no return.